



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Title: Automated Evaluation of Descriptive Answer Sheets Using OCR and Semantic NLP

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Guide Details

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Mini Project Outlines

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Introduction

The **AI Answer Sheet Evaluator** is an automated system designed to reduce the effort and time required for manual correction of answer sheets. Traditional evaluation methods are slow and may lead to inconsistencies. This project uses **OCR (Optical Character Recognition)** to extract handwritten text and **NLP (Natural Language Processing)** to understand and evaluate the answers. It can process answer sheets captured through mobile photos and automatically generate marks with feedback.

- Manual correction is time-consuming
- Chances of human errors and bias
- Automated system for fast evaluation
- Uses OCR to read handwritten text
- Uses NLP to understand answers
- Generates marks and feedback automatically

Existing System Vs Proposed

● Existing System	● Proposed System
Manual correction by teachers	Fully automated grading using AI
Time-consuming, especially for large classes	Faster evaluation (within seconds)
High chances of human errors and bias	More accurate and consistent results
Inconsistent marking between evaluators	Standardized and fair evaluation
Difficult to manage and scale	Scalable for large number of students
No support for handwritten answer automation	Accepts handwritten answer sheets (images)
No feedback or performance analysis	Provides marks, feedback, and confidence score
No intelligent understanding	Uses OCR to extract text
—	Uses NLP to understand answer meaning

Traditional answer sheet evaluation depends on manual correction, which is slow and may lead to errors or inconsistency. The proposed system uses AI-based automation to evaluate handwritten answers quickly, accurately, and fairly, while also providing useful feedback.

👉 **Simple Meaning:**

Old System = Slow & Manual

New System = Fast, Smart & Automatic

Research Gaps

Although many automated grading systems exist, they still have several limitations, especially when dealing with real-world handwritten answer sheets. These gaps highlight the need for a more advanced and reliable solution.

- Most systems are designed for **typed or digital text only**
- OCR tools struggle with **handwritten text variations**
- Poor accuracy with **cursive or unclear handwriting**
- No proper system for handling **multi-question answer sheets**
- Difficulty in **segmenting answers question-wise**
- Lack of **feedback generation for students**
- No proper **accuracy or confidence checking mechanism**
- Poor handling of **low-quality images (blur, lighting issues, noise)**
- Limited ability to **understand meaning (semantic understanding)**
- Many systems rely only on **keyword matching (not context)**

Problem Statement

In the current education system, evaluating answer sheets manually is a time-consuming and challenging task. Teachers have to handle a large number of papers, which can lead to delays, inconsistency, and errors in grading. Existing automated tools are not capable of accurately handling handwritten answers or understanding their meaning, creating a need for a smarter solution.

- Manual evaluation is **slow and time-consuming**
- Difficult to manage **large number of answer sheets**
- High chances of **human errors and bias**
- Inconsistent grading between different evaluators
- Heavy workload on teachers
- Existing systems fail to handle **handwritten answers properly**
- Lack of **intelligent understanding of answers**

Proposed Project Objectives

The main objective of this project is to develop an intelligent system that can automatically evaluate handwritten answer sheets by reading, understanding, and comparing them with model answers to generate accurate results.

Objectives

- Develop an **OCR system** to read handwritten answers
- Use **NLP techniques** to understand answer meaning
- Compare **student answers with model answers**
- Generate: Marks , Feedback , Confidence score
- Ensure **fair and unbiased evaluation**
- Achieve accuracy close to **human grading (85–95%)**

👉 **Simple Meaning:** Make a system that can **read + understand + give marks like a teacher**

Proposed Project Outcomes

The project results in a smart automated system that improves the efficiency and quality of answer sheet evaluation while reducing manual effort.

Outcomes

- Fully **automatic grading system**
 - Reduces **teacher workload significantly**
 - Provides **fast evaluation (within seconds)**
 - Ensures **consistent and fair marking**
 - Handles **large number of answer sheets easily**
 - Generates **accurate results with feedback**
 - Suitable for **real-world academic use**
- 👉 **Simple Meaning:** System makes grading **fast, easy, and accurate**

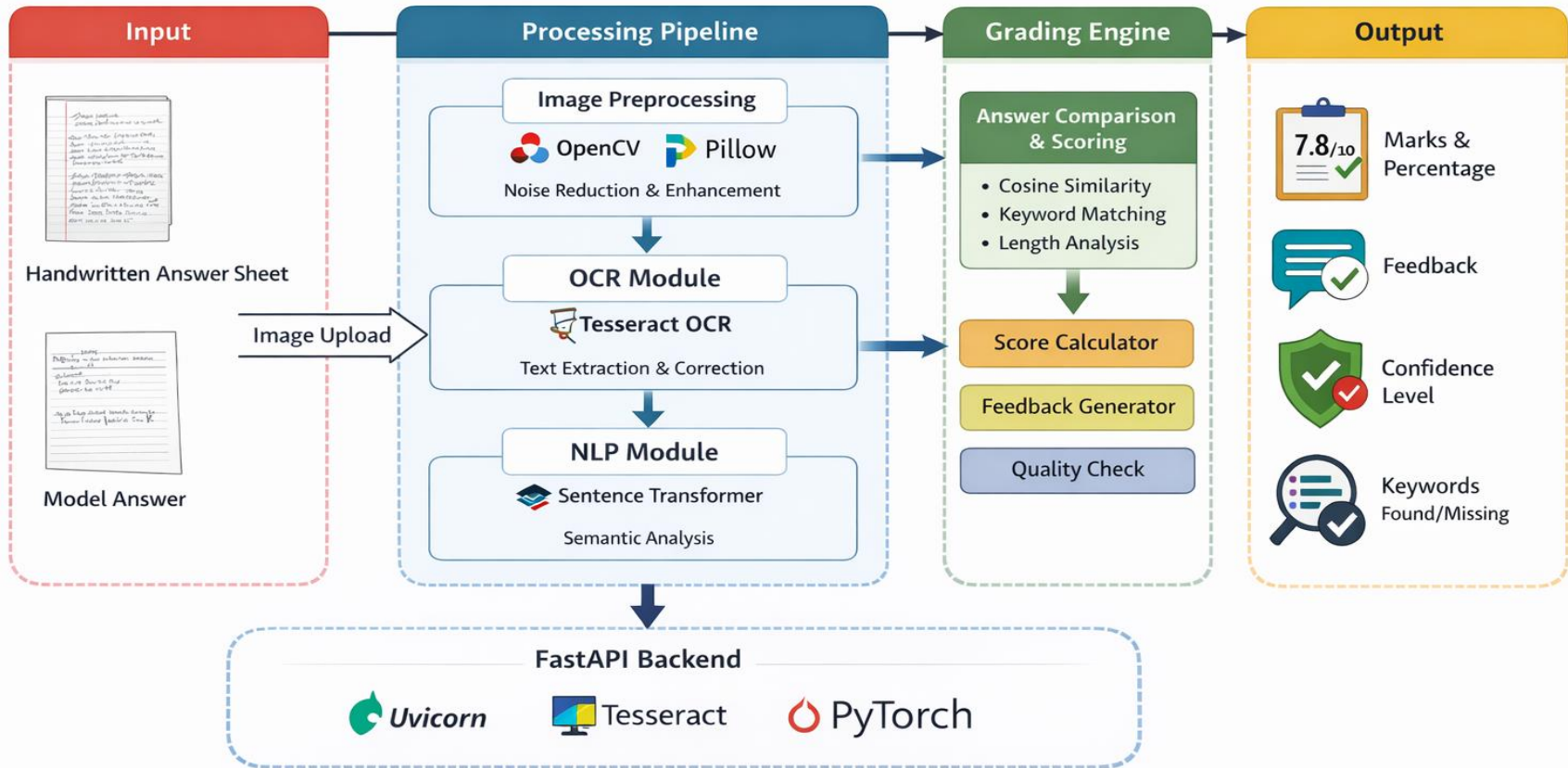
Software and Hardware Requirements

 Hardware Requirements	 Software Requirements
Processor: Intel i5 or higher	Programming Language: Python 3.x
RAM: Minimum 8 GB	Backend: FastAPI + Uvicorn
Storage: At least 5 GB free space	OCR Engine: Tesseract OCR
GPU: Optional (for faster processing)	Image Processing: OpenCV + Pillow
—	NLP Model: Sentence Transformers
—	Libraries: NumPy, Transformers, PyTorch

Hardware runs the system, software performs the smart processing

Architectural Diagram

AI Answer Sheet Evaluator System Architecture



PROPOSED METHOD AND ALGORITHMS

The system follows a structured pipeline that converts handwritten answer sheets into meaningful scores using image processing, OCR, and NLP techniques.

Step 1: Image Processing

- Improve image quality
- Remove noise

Step 2: OCR

- Extract text from image

Step 3: NLP Processing

- Understand meaning of text

Step 4: Evaluation

- Compare with model answer

Algorithms Used

The system uses multiple algorithms to ensure accurate text extraction and fair evaluation.

1. Image Preprocessing

- Noise removal
- Contrast enhancement

2. OCR Quality Check

- Checks if extracted text is valid

3. Grading Algorithm

Weighted Scoring Formula:

- 50% Semantic Similarity
- 25% Sentence Coverage
- 20% Keyword Coverage
- 5% Length Ratio
- 👉 **Simple Meaning:**

Image → Clean → Read → Understand → Compare → Marks

Why This Formula?

- Ensures conceptual understanding
- Prevents keyword stuffing
- Avoids very short answers

Schema

◆ System Stages	◆ Details / Logic (with Formula & Techniques)
Input	Student Answer (Image) + Model Answer (Text/Image)
Image Preprocessing	$g' = 255 - G$ (green channel), Resize (~3000px), CLAHE, Otsu threshold, Noise removal
OCR Module	Tesseract OCR (PSM 4 & 6), Best output selection, 180+ text corrections
OCR Quality Check	$q = \text{valid_words} / \text{total_words} \rightarrow \text{Good} \geq 0.55, \text{Poor} < 0.35$
NLP Processing	Tokenization, Sentence Embeddings (Sentence Transformer), TF-IDF keywords
Grading Algorithm	$\text{Score} = 0.5 \times \text{Similarity} + 0.25 \times \text{Coverage} + 0.20 \times \text{Keywords} + 0.05 \times \text{Length}$
Output	Marks, Percentage, Feedback, Confidence, Keywords Found/Missing
Backend System	FastAPI + Uvicorn + PyTorch (handles processing & APIs)

Overall Project Schema (Technical)



Input



Student Answer (Image) + Model Answer (Text/Image)



Image Preprocessing

- $g' = 255 - g$ (green channel)
- Resize (~3000px), CLAHE, Otsu threshold
- Noise removal



OCR Module

- Tesseract OCR (PSM4+ & 6)
- Best output selection
- 180+ text corrections
- 180+ text corrections



OCR Quality Check

$q = \text{valid_words} / \text{total_words} \rightarrow \text{Good} \geq 0.55, \text{Poor} < 0.35$



NLP Processing

- Tokenization
 - Sentence Embeddings
 - TF-IDF keywords
- $\text{Score} = 0.5 \times \text{Similarity} + 0.25 \times \text{Coverage} + 0.20 \times \text{Keywords}$
- $q \geq 0.55 \rightarrow \text{Good}, q = 0.55 \rightarrow \text{Poor (warning/reject)}$



Grading Algorithm

- Tokenization
- Sentence embeddings (Sentence Transformer)
- TF-IDF keywords
- TF-IDF keywords



Backend System

Uvicorn + FastAPI + Tesseract + PyTorch (handles processing 3 APIs)

Evaluation Schema (Scoring Logic)



$\text{Final Score} = 0.50 \times \text{Semantic Similarity}$	0.50%
$+ 0.25 \times \text{Sentence Coverage}$	0.25%
$+ 0.20 \times \text{Keyword Matching}$	0.20%
$+ 0.05 \times \text{Answer Length}$	0.05%

Total = **1.00 (100%)**



Student Answer + Model Answer

Student Answer (image)
Model Answer (Text/Image)



Semantic Similarity (0.50)



Sentence Coverage (0.25)



Keyword Matching (0.20)



Answer Length Check (0.05)



Final Score Calculation



Marks + Feedback

Weight Distribution

Semantic Meaning	50%
Sentence Coverage	25%
Keywords Match	20%
Answer Length	5%
Total	100%

Weight Distribution

Semantic Meaning	50%
Sentence Coverage	25%
Keywords Match	20%
Answer Length	5%

Results

The system shows strong performance in evaluating handwritten answers, achieving results close to human grading while maintaining speed and consistency.

Results

- Accuracy: **~89% (close to human evaluation)**
- Correlation with human grading: **85–95%**
- Processing time: **2–5 seconds per answer sheet**
- Works well for **clear and moderately neat handwriting**

Output Provided

- Marks and percentage
- Feedback (strengths & improvements)
- Confidence level
- Keywords found and missing

 **Simple Meaning:** System gives **fast and accurate results like a teacher**

Research Paper (Acceptance Proof)



Sathvik Rudam <rudamsathvik@gmail.com>

3rd International Conference on Artificial Intelligence Trends and Pattern Recognition, 2026 : Submission (105) has been edited.

2 messages

Microsoft CMT <noreply@msr-cmt.org>
To: rudamsathvik@gmail.com

Tue, Mar 10, 2026 at 2:06 PM

Hello,

The following submission has been edited.

Track Name: Advances in Pattern Recognition: Techniques and Applications

Paper ID: 105

Paper Title: Automated Evaluation of Descriptive Answer Sheets Using OCR and Semantic NLP

Abstract:

Grading handwritten answer sheets can be tiring, inconsistent, and tough to handle, especially in large schools. This research introduces an AI-based system for evaluating answer sheets. It aims to simplify and automate the grading process using Optical Character Recognition (OCR) and Natural Language Processing (NLP) technologies. The system extracts handwritten text from scanned papers with Tesseract OCR, using adaptive image preprocessing methods to improve recognition performance. After conversion to digital form, the text is analyzed with transformer-based Sentence Transformer models to grasp the meaning of students' responses. Instead of just matching keywords, the system measures the similarity between student answers and model solutions to give accurate and meaningful scores. A quality control feature spots unclear or low-confidence text to lessen the chance of wrong evaluations. Built with a FastAPI backend, the platform offers a user-friendly web interface. Educators can upload answer sheets and quickly receive results with confidence scores. By combining reliable OCR, understanding of language context, and automated scoring, this system provides a scalable, clear, and efficient solution for modern academic grading.

Created on: Tue, 10 Mar 2026 08:34:20 GMT

Last Modified: Tue, 10 Mar 2026 08:36:52 GMT

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Secondary Subject Areas: Not Entered

Submission Files:

paper.pdf (5 Mb, Tue, 10 Mar 2026 08:33:52 GMT)

Submission Questions Response: Not Entered

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International Conference on Recent Advancement in Electrical, Computer and Communication Technologies (IECCT 2026) : Submission (1010) has been edited.

1 message

Microsoft CMT <noreply@msr-cmt.org>
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Tue, Mar 10, 2026 at 2:10 PM

Hello,

The following submission has been edited.

Track Name: IECCT2026

Paper ID: 1010

Paper Title: Automated Evaluation of Descriptive Answer Sheets Using OCR and Semantic NLP

Abstract:

Grading handwritten answer sheets can be tiring, inconsistent, and tough to handle, especially in large schools. This research introduces an AI-based system for evaluating answer sheets. It aims to simplify and automate the grading process using Optical Character Recognition (OCR) and Natural Language Processing (NLP) technologies. The system extracts handwritten text from scanned papers with Tesseract OCR, using adaptive image preprocessing methods to improve recognition performance. After conversion to digital form, the text is analyzed with transformer-based Sentence Transformer models to grasp the meaning of students' responses. Instead of just matching keywords, the system measures the similarity between student answers and model solutions to give accurate and meaningful scores. A quality control feature spots unclear or low-confidence text to lessen the chance of wrong evaluations. Built with a FastAPI backend, the platform offers a user-friendly web interface. Educators can upload answer sheets and quickly receive results with confidence scores. By combining reliable OCR, understanding of language context, and automated scoring, this system provides a scalable, clear, and efficient solution for modern academic grading.

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Submission Files:

paper.pdf (5 Mb, Tue, 10 Mar 2026 08:40:27 GMT)

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Conclusion

The project successfully demonstrates an AI-based solution for automating answer sheet evaluation, improving efficiency and fairness in the grading process.

- Successfully **automates answer sheet grading**
- Reduces **time and manual effort** for teachers
- Minimizes **human bias and errors**
- Provides **consistent and fair evaluation**
- Works effectively with **handwritten answers**
- Uses **OCR + NLP for intelligent grading**
- Suitable for **modern education systems**

System makes grading **easy, fast, and fair**

Thank You